

HOULTON, ME, USA

WMO: 727033

Lat: 46.119N

Lon: 67.793W

Elev: 476

StdP: 14.44

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14609

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.6	-8.9	-22.5	1.6	-12.7	-17.3	2.2	-7.3	22.5	23.2	20.5	20.8	4.1	200	0.605

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.8	85.2	69.6	82.1	67.7	79.2	66.0	72.2	80.8	70.4	78.1	68.8	75.7	9.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.5	110.5	76.0	67.8	104.0	74.1	66.1	97.9	72.6	36.1	80.7	34.6	78.2	33.2	75.6	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.8	18.1	16.4	-23.6	90.4	5.9	2.6	-27.8	92.3	-31.3	93.8	-34.6	95.2	-38.9	97.0		
(5)			-23.6	75.7	5.5	1.5	-27.6	76.8	-30.8	77.6	-33.9	78.5	-37.8	79.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.2	13.3	15.7	25.9	38.8	51.4	60.4	66.5	64.4	56.5	44.8	33.3	21.2		
	DBStd	20.38	12.91	11.77	10.45	7.52	7.24	6.93	5.42	5.66	7.65	7.85	9.40	11.08		
	HDD50	4884	1137	961	748	344	69	3	0	0	23	201	506	892		
	HDD65	8914	1602	1381	1212	786	427	173	46	80	272	626	952	1357		
	CDD50	1661	0	0	1	8	112	316	512	448	219	41	4	0		
	CDD65	215	0	0	0	0	5	36	93	63	18	0	0	0		
	CDH74	2184	0	0	1	4	109	406	893	610	156	5	0	0		
	CDH80	506	0	0	0	0	23	98	209	140	36	0	0	0		
(14) Wind	WSAvg	7.1	7.7	7.8	8.4	8.1	7.5	6.6	6.1	5.7	6.1	6.9	7.4	7.4		
(15) Precipitation	PrecAvg	38.7	2.7	2.2	2.5	2.6	3.2	3.4	3.6	4.0	3.6	3.5	3.8	3.4		
	PrecMax	56.1	6.0	3.9	4.9	6.1	6.3	9.2	6.7	9.8	9.8	7.9	8.5	5.8		
	PrecMin	21.7	0.2	0.4	0.4	0.4	0.4	1.2	1.6	0.5	1.3	1.1	1.7	1.0		
	PrecStd	6.6	1.5	0.9	1.0	1.3	1.5	1.8	1.3	2.1	1.8	1.9	1.6	1.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.1	47.4	56.8	70.0	83.1	87.5	88.7	88.4	84.8	73.6	61.7	51.5		
		MCWB	48.9	43.8	45.9	51.5	64.5	71.3	71.6	70.7	70.7	61.3	58.6	49.3		
	2%	DB	42.3	41.2	48.0	62.6	76.7	82.6	84.9	83.8	78.5	66.6	56.6	45.0		
		MCWB	40.1	37.5	40.4	48.9	59.2	67.4	70.1	69.1	67.3	58.5	52.7	42.8		
	5%	DB	36.6	37.1	44.0	57.2	71.5	78.8	82.1	80.3	73.7	62.7	51.9	39.9		
		MCWB	34.4	34.1	38.3	47.0	56.3	65.0	68.0	67.0	64.2	57.0	48.4	37.7		
	10%	DB	32.3	33.9	40.3	52.6	66.6	74.8	79.2	77.1	70.3	58.8	47.5	36.3		
		MCWB	30.6	31.6	35.3	44.3	54.5	62.8	66.7	65.5	62.8	53.3	44.4	34.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.5	45.4	48.6	57.4	67.4	73.6	74.9	73.8	72.6	64.7	59.9	49.5		
		MCDB	50.4	46.4	52.4	64.2	78.6	84.3	83.6	82.6	81.6	67.4	61.7	50.7		
	2%	WB	40.3	38.7	43.0	52.6	62.3	70.3	72.3	71.4	69.3	61.1	54.2	43.2		
		MCDB	42.1	40.5	46.1	57.7	71.1	78.6	80.9	79.3	74.2	65.5	55.8	44.7		
	5%	WB	34.4	34.4	38.7	48.6	59.4	67.9	70.5	69.6	66.9	57.6	48.6	38.2		
		MCDB	36.2	36.5	43.0	55.6	67.2	75.1	78.5	76.5	72.0	62.7	50.9	39.7		
	10%	WB	30.6	31.8	35.9	45.0	56.6	65.3	68.8	67.8	64.1	53.7	44.4	34.3		
		MCDB	32.3	33.9	39.6	51.6	64.8	71.9	76.0	74.4	68.7	57.9	47.3	35.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.2	20.8	19.3	19.6	22.6	22.2	21.8	22.7	22.3	19.0	14.9	16.0		
		MCDBR	21.0	19.0	22.7	29.5	32.0	27.9	25.3	26.7	25.2	24.4	19.8	17.2		
	5% WB	MCWBR	20.5	16.2	16.8	18.2	17.2	15.0	12.4	13.6	15.1	16.8	17.1	16.3		
		MCWBR	21.4	17.5	20.1	24.2	24.8	23.0	21.2	21.3	21.1	21.1	18.0	17.2		
(40) Clear-Sky Solar Irradiance	taub		0.259	0.278	0.310	0.342	0.370	0.391	0.416	0.394	0.364	0.322	0.300	0.278		
		taud	2.426	2.338	2.366	2.408	2.377	2.317	2.271	2.336	2.400	2.524	2.536	2.453		
	Ebn at Noon		273	287	292	291	285	278	269	271	270	268	255	251		
		Edh at Noon	22	29	33	35	37	40	41	37	32	24	20	19		
(44) All-Sky Solar Radiation	RadAvg		465	745	1115	1398	1586	1717	1735	1536	1236	754	441	340		
	RadStd		53	74	75	131	179	123	141	96	91	70	50	31		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	+2.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (22 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page